A Model-Driven Deep Mixture Network for Robust Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral anomaly detection (HAD) aims to identify samples with unknown atypical spectra from the background. Deep learning (DL)-based methods, particularly autoencoders (AEs), have proven effective in uncovering the underlying profiles for HAD. However, in real-world applications of hyperspectral images (HSIs), complex background land covers and anomaly corruptions are common, leading to two issues: 1) a low-dimensional manifold characterized by DL-based HAD methods can only reveal a few underlying variation factors of the background distribution and cannot capture the complex structures behind land covers of all categories and 2) DL-based HAD methods trained on anomaly-contaminated HSIs tend to overfit specific anomalies, resulting in poor background characterization. To tackle these issues, this study presents a novel and robust framework for HAD called model-driven deep mixture network (MDMN) that combines the strengths of model-driven and data-driven approaches while emphasizing interpretability. By assuming that the background, consisting of various land-covers, arises from a mixture of low-dimensional manifolds, the MDMN incorporates a novel deep mixture module to comprehensively characterize the background. This module uses a low-dimensional manifold learned by an AE to represent a specific category of background land covers. To mitigate the impact of anomaly corruptions, the MDMN incorporates a convex relaxation of a sparse constraint, which helps prevent overfitting anomalies. Extensive experimental results demonstrate that the proposed MDMN offers more satisfactory and robust detection performance.**

***Index Terms*— Hyperspectral anomaly detection (HAD), manifold learning, mixture network, model-driven.**

I. Introduction

HYPERSPECTRAL remote sensing, which provides detailed spectral information ranging from visible to infrared, plays a pivotal role in Earth observation and deep space exploration [1], [2]. Building on top of rich spectral and spatial information, a series of hyperspectral image (HSI) interpretation techniques have been investigated [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14]. One of the interesting HSI interpretation techniques is hyperspectral anomaly detection (HAD) [15], [16], [17], [18], [19], which aims to identify unique spectra that deviate from the surrounding background in an unsupervised manner and dig out the inherent profiles of the HSIs that convey valuable data information [20]. Consequently, HAD is considered as an essential step in critical decision-making systems such as environmental monitoring and military camouflage detection, enabling curiosity-driven exploration of unknown events and environments [21], [22].

In the past years, various methods have been proposed for HAD. Many works focus on exploring data or physical properties of HSI background and anomalies and introduce various prior structures to regularize and separate them [15]. Along this research line, the representative methods include background priors with Gaussian distribution model [23], [24], [25], Gaussian mixture model [26], [27], [28], sparsity representation [29], [30], [31], [32], [33], collaborative representation (CR) [34], [35], [36], [37], low-rank representation [28], [37], [38], [39], [40], [41], piecewise smooth [42], specific topological structure [43], and spectral mixing model [27], [39]. Especially, inspired by the fact that anomalies sparsely appear at different locations over an HSI with random patterns such as shape, scale, and spectral curves, researchers represented this configuration of anomalies by sparsity prior [38] and its variants, i.e., group sparsity prior [41] and structured sparsity prior [44], and mixture of Gaussian (MoG) distribution [28]. These methods have achieved state-of-the-art performance when the background can be well-represented. The rationality of these methods depends on the assumed subjective priors imposed on the unknown background and anomalies. In real scenarios, however, these methods could not always adapt to HSIs with complex and diverse image contents collected from different scenes [15], [16].

Witnessing the great strides of deep learning (DL) in a wide range of applications, increasing efforts have been devoted to the design of DL-based HAD methods [15], [45]. The current DL-based HAD methods mainly focus on designing network architectures and training strategies, and then train network parameters based on the HSIs to be processed to reconstruct background spectra and extract the anomalies. Typical DL-based HAD methods include autoencoder (AE) [46]-based methods [47], [48], [49], [50], [51], [52], [53], [54], the methods [55], [56], [57], [58], [59], [60], [61] based on the

Manuscript received 4 March 2023; revised 24 June 2023 and 30 July 2023; accepted 17 August 2023. Date of publication 30 August 2023; date of current version 14 September 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62121001 and Grant 62071360 and in part by the Fundamental Research Funds for the Central Universities and the Innovation Fund of Xidian University under Grant 5001-20109215456. *(Corresponding author: Kai Jiang.)*

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Digital Object Identifier 10.1109/TGRS.2023.3309960

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combination of generative adversarial network (GAN) [62] and AE, and some others [63], [64], [65].

The DL-based HAD methods conduct representation learning in the latent space learned by AEs, but possess evident deficiencies. For HSIs with complex structures and spectral variability [10] caused by the variations in environmental conditions, the performance of DL-based HAD methods is far from satisfactory. The primary drawback of the existing DL-based methods is their reliance on a single low-dimensional manifold, learned by an AE, to represent diverse land-cover categories in HSIs. A single AE that characterizes a low-dimensional manifold tends to have a simplistic structure, limiting its ability to capture the inherent variability present in all land-cover categories in HSIs. This deficiency principally includes the failure to capture endmember spectral variability [66]. Consequently, for HSIs, the latent representation that is entangled with several causes of endmember variability is poorly characterized. Moreover, the majority of current DL-based HAD methods primarily focus on learning a representation model for background spectra. However, this approach becomes problematic when the training set contains a small proportion of anomalies, leading to overfitting on anomalous samples and yielding significant reconstruction errors for background spectra. The application of AE to the SanDiego dataset, which exhibits diverse background structures and anomaly corruption, reveals its limited performance in both background representation and anomaly detection, as depicted in Fig. 1. Fig. 1(b)–(d) depicts the significant reconstruction errors in the background spectra reconstructed by an AE trained on the SanDiego dataset. Conversely, Fig. 1(a) demonstrates that the AE's learned low-dimensional manifold is capable of effectively representing anomalies, i.e., the aircraft, resulting in minimal reconstruction error values for aircraft spectra. Thus, training a single AE to characterize the diverse background spectra collected from various land covers in practical scenarios is deemed inappropriate. Instead, the primary focus should be on acquiring proper representations by leveraging the underlying structure of HSIs and determining a suitable representation model capable of discerning anomalous spectra while preserving the background structures. Furthermore, DL-based HAD methods suffer from a notable limitation in terms of interpretability. They are often viewed as black-box detectors that lack insight into their plausibility, disregarding the inherent priors of HSI such as the spectral mixing [67], spectral variability [10], [66], and the sparsity of anomalies [38]. Therefore, it is essential to interpret the decisions made by DL-based methods in HAD to enable researchers and users to comprehend, trust, and enhance these decisions [68], [69].

The above shortcomings prompt the question, particularly for practical applications, whether it is possible to establish a comprehensive DL-based model from an interpretable perspective for HAD with respect to dataset with complicated and diverse structures. Certainly, a more promising approach is to thoroughly investigate the inherent structures of both background and anomalies and endeavor to integrate this prior knowledge into the learning process. This integration aims to leverage the benefits of the traditional model-driven prior-based methods and the current data-driven DL-based approaches. Drawing from this insight, we design a robust HAD depending on the following three critical attributes: 1) the complexity and variability of the background spectra distribution in the latent space, which limits the effectiveness of a single low-dimensional manifold in capturing this intricate structure; 2) the deviation of anomalies from the underlying model constructed on the low-dimensional manifolds of background spectra; and 3) the significantly lower occurrence probability of anomalies compared with normal pixels, which aligns with the sparse prior. Capitalizing on these attributes, we introduce the model-driven deep mixture network (MDMN) as an interpretable approach to HAD. Unlike previous models, the MDMN takes into account the characteristics of anomalies and background endmember spectral variability, combining the strengths of model-driven and data-driven frameworks to address the HAD task. Specifically, the background distribution, which exhibits spectral variability and diverse structures, is represented as a mixture of manifolds. Each manifold, learned by an AE, corresponds to a specific background endmember. Notably, the membership assignment, i.e., the abundance, within a manifold is enforced to be sparse. This encourages the dominant characterization of the variability within a specific background class, while others contribute to its overall representation. Given the sparse nature of anomalies, we enhance their prominence through a convex relaxation of the sparse operator. This helps eliminate anomaly corruption and prevents model overfitting anomalies. Experiments conducted on various datasets demonstrate the superior performance of the proposed MDMN. The main contributions of this article are summarized as follows.

1) We introduce a novel MDMN framework for HAD from an interpretable perspective, which is the first of its kind to optimize both the model-driven and data-driven approaches simultaneously.
2) We develop a novel deep mixture module that demonstrates superior performance in modeling background spectra with spectral variability and effectively captures deviations associated with anomalies.
3) To mitigate the effects of anomaly corruption and obtain an optimal solution for MDMN, we apply a convex relaxation technique to the sparse constraint on the anomaly matrix. Extensive experiments validate the effectiveness, robustness, and interpretability of MDMN

The remainder of this article is organized as follows. Section III provides a detailed description of the proposed HAD framework. Section IV presents the experimental results and analysis. Finally, our conclusions are drawn in Section V.

II. Related Works

This section provides a brief review on the related works of HAD, which can be categorized into three groups: 1) statistics-based; 2) reconstruction-based; and 3) DL-based [15].

A. Statistics-Based HAD

The Reed and Xiaoli (RX) algorithm [23], which is regarded as one of the most well-known traditional statistics-based

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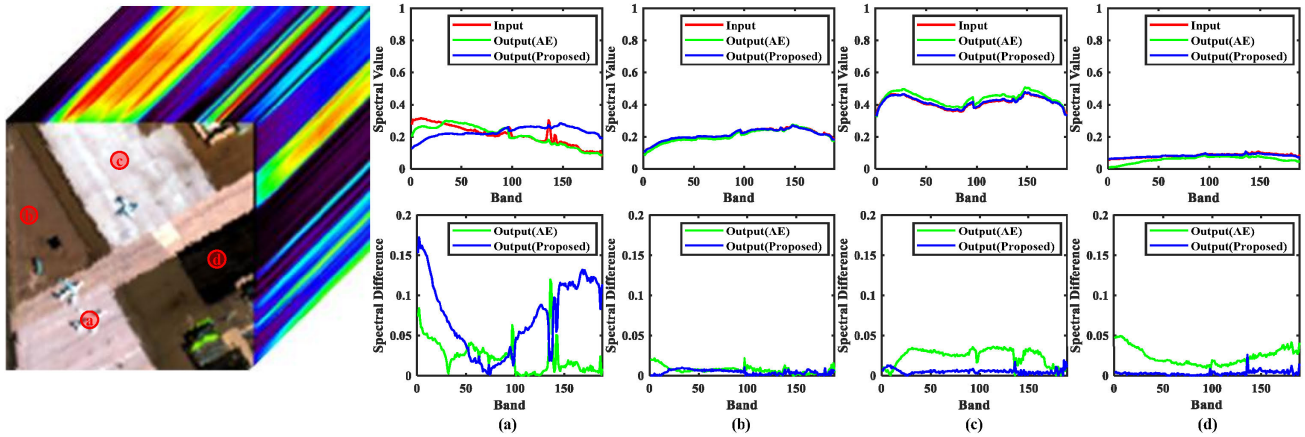


Fig. 1. Performance comparison between AE and the proposed MDMN. (a) Shows an anomaly while the others [i.e., (b), (c), and (d)] show backgrounds. (Top) Input spectra, output of an AE, and output produced by MDMN. (Bottom) The spectral difference between input spectra and the output of the models. Both the models are trained on the SanDiego dataset, which contains complex backgrounds contaminated by anomalous spectra. While the AE aims to minimize reconstruction errors across the entire dataset, it exhibits subpar reconstruction performance for diverse background spectra but relatively better reconstruction for anomalous spectra. In contrast, the proposed MDMN prevents anomaly corruption and enforces the latent representation of samples to align with their dominant background endmembers. As a result, diverse background spectra yield low reconstruction errors, while anomalous spectra result in high reconstruction errors.

methods, is essentially based on the multivariate Gaussian distribution assumption for background spectra. Improved versions of RX, such as LRX [24] that uses local statistics and kernel-RX [25] benefiting from kernelization, suffer from a high computational burden. The cluster-based anomaly detection (CBAD) [26] method uses the Gaussian mixture model to present a robust representation of background statistics. Analogously, a method for HAD that combines spectral unmixing [10], [67], [70] and Gaussian mixture models is proposed in [27], However, due to the strong assumption on background land-cover distribution, the performance of the statistics-based methods is limited when considering HSIs with complicated scenes [15], [16].

B. Reconstruction-Based HAD

The advancement of compressed sensing theory [71] has led to the emergence of reconstruction-based HAD methods, which can be categorized into three subcategories: sparse representation (SR)-based methods, CR-based methods, and low-rank-representation (LRR)-based methods. Several SR-based HAD methods [29], [30], [31], [32], [33] have been proposed by leveraging the assumption that spectra can be effectively represented using a limited number of atoms from an overcomplete dictionary, while anomalies cannot. These methods primarily focus on dictionary construction [29], [30] and the optimization of the SR models [31], [32]. In contrast to SR-based methods, CR-based methods [34], [35], [36] emphasize the collaborative nature of all the dictionary atoms and aim to use the entire dictionary for linearly representing normal spectra, resulting in improved performance. The LRR-based methods place greater emphasis on the global structure of HSIs, instead of solely focusing on the spectrum-by-spectrum relationship. Several methods have been proposed [37], [38], [41], [42], [72] that extensively investigate the low-rank prior of background and the sparsity prior of anomalies for detecting anomalies in HSIs. Huang et al. [39] and Qu et al. [40] enhance the LRR model by incorporating spectral mixing prior to improve background representation. Li et al. [28] constructed a detector that further models the anomaly component as an MoG. Differently, Chang et al. [33] introduce an iterative process for an anomaly detector, incorporating spectral and spatial prior knowledge from previous iterations, thereby enhancing the detection performance of the anomaly detector but exhibiting time-consuming iterative computations. In principle, the above subjective image priors, such as low rank, MoG, piecewise smooth, and sparsity, are hand-crafted and could not always be robust enough for HSIs in real applications [73], [74].

C. DL-Based HAD

With the remarkable advancements of DL in various applications, there has been a growing focus on developing DL-based methods for HAD, which can be categorized as supervised or unsupervised, depending on whether sample support is needed [15], [45]. The typical supervised methods in HAD [63], [64] are based on convolutional neural network (CNN), which are trained on labeled reference dataset and transferred to HAD task. Typical unsupervised DL-based HAD methods are based on AE and the combination of GAN [62] and AE. The AE-based methods [47], [48], [49], [50], [51], [52], [53], [54], [65] applied in HAD can construct abstract, hierarchical, and high-level representations for background spectra and have demonstrated very impressive HAD performance. Particularly, the robust graph AE (RGAE) [53] performs background reconstruction and anomaly detection with regularization derived from the l21-norm and superpixel segmentation-based graphs. Several methods [55], [56], [57], [58], [59], [60], [61] use a combination of GANs and AEs to create an adversarial learning framework. This framework involves adding discriminators to AEs, with the objective of extracting intrinsic and discriminative spectral features to improve the discrimination between anomalies and

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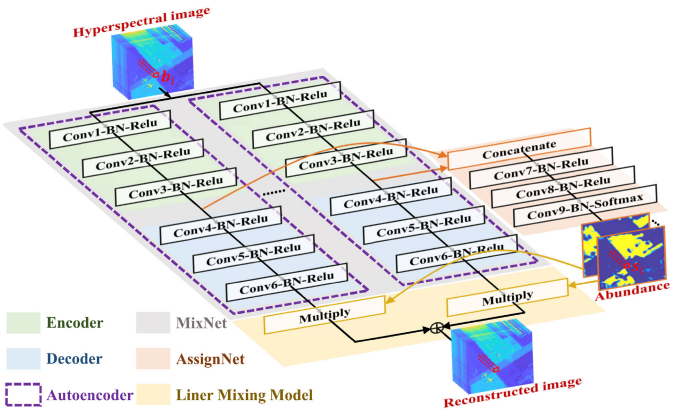


Fig. 2. Network architecture of the proposed MDMN.

background. However, these DL-based methods have inherent limitations, including overfitting to anomalies, insufficient representation learning capability, and limited interpretability, among others.

III. Proposed HAD Framework

A. Problem Formulation

In significant contrast to normal samples, anomalies in HSIs are not labeled, which makes the traditional expert-based supervised methods weak or prone to errors. Besides, the normal background spectra with complex structure, which are assumed to be generated from a collection of low-dimensional nonlinear manifolds, pose a challenge to HAD. The proposed MDMN infers anomalies based on the motivations that they present a sparse property, while normal background spectra with complex structure can be clustered into K categories, in which each category is sampled from a certain different low-dimensional manifold.

For an observed unlabeled HSI denoted as $\mathbf{X} \in \mathbb{R}^{H\times W\times L}$, where H, W, and L represent the height, the width, and the band number of $\mathbf{X}$, respectively, it can be unfolded as

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N] \in \mathbb{R}^{L\times N} \tag{1}$$

where $N = H \times W$ and $\mathbf{x}_i (i = 1, 2, \ldots, N)$ is the ith spectrum contained in the Euclidean space $\mathbb{R}^L$. It can be rationally separated as

$$\mathbf{X} = \mathbf{B} + \mathbf{A} \tag{2}$$

where $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_N]$ and $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_N]$ represent the background component and anomaly component of $\mathbf{X}$, respectively. Then, the aim of most DL-based anomaly detection methods is to estimate the mapping function (formulated by a neural network) from $\mathbf{X}$ to $\mathbf{B}$ (or $\mathbf{A}$).

Instead of constructing a complex DL architecture heuristically, we first consider the conventional prior-based methodology through exploiting the prior structures for representing background component and anomaly component. Specifically, by adopting the linear spectral mixing model [70] with abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC), the ith spectrum $\mathbf{b}_i$ of the background component can be modeled as

$$\mathbf{b}_i = \mathbf{E}\mathbf{s}_i + \mathbf{n}_i \in \mathbb{R}^{L\times 1}$$
$$\text{s.t.} \begin{cases} \mathbf{s}_i \geq \mathbf{0} \\ \mathbf{1}^T\mathbf{s}_i = 1 \end{cases} \tag{3}$$

where $\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_K] \in \mathbb{R}^{L\times K}$ is a collection of K endmembers, $\mathbf{s}_i$ represents the abundance vector for $\mathbf{b}_i$ with respect to the endmember matrix $\mathbf{E}$ that sums to 1, and $\mathbf{S} = [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_N] \in \mathbb{R}^{K\times N}$ is the abundance matrix. Besides, the ith row vector of $\mathbf{S}$ represents the ith vectorized abundance map which indicates the locations where the corresponding background endmembers repeatedly appear. Here, $\mathbf{n}_i$ is the ith column vector in noise component $\mathbf{N} \in \mathbb{R}^{L\times N}$. Thus, model (2) can be reformulated as

$$\mathbf{X} = \mathbf{E}\mathbf{S} + \mathbf{N} + \mathbf{A}. \tag{4}$$

Therefore, for predicting the anomaly component from the unlabeled HSI, the key issue is to calculate $\mathbf{ES}$ and $\mathbf{A}$ from $\mathbf{X}$ alternately, and the corresponding optimization problem is

$$\min_{\mathbf{E},\mathbf{S},\mathbf{A}} ||\mathbf{B} - \mathbf{E}\mathbf{S}||_F + \lambda_1 g_1(\mathbf{S}) + \lambda_2 g_2(\mathbf{A})$$
$$\text{s.t.} \begin{cases} \mathbf{X} = \mathbf{B} + \mathbf{A} \\ \mathbf{S} \geq \mathbf{0} \\ \mathbf{1}_K^T\mathbf{S} = \mathbf{1}_N \end{cases} \tag{5}$$

where λ_1 and λ_2 are the tradeoff hyperparameters, and $g_1(\mathbf{S}) = \sum_i^N ||\mathbf{s}_i||_1$ and $g_2(\mathbf{A}) = ||\mathbf{A}||_{2,1}$ are the regularizers to deliver the prior structures of normal background component $\mathbf{B}$ and a convex relaxation of the sparse constraint on the anomaly component $\mathbf{A}$, respectively. Equation (5) can be further derived as

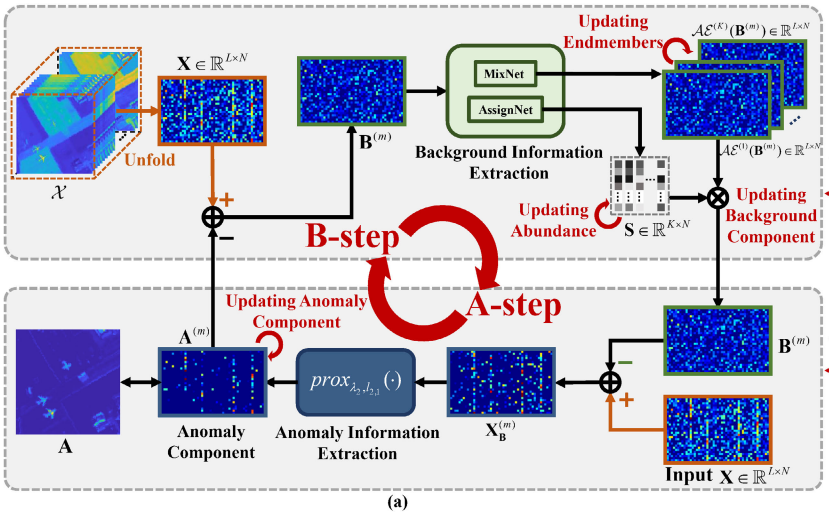
$$\min_{\mathbf{E},\mathbf{S},\mathbf{A}} \sum_{i=1}^{N} \left[||\mathbf{b}_i - \mathbf{E}\mathbf{s}_i||_2^2 + \lambda_1 \|\mathbf{s}_i\|_1\right] + \lambda_2 \|\mathbf{A}\|_{2,1}$$
$$\text{s.t.} \begin{cases} \mathbf{X} = \mathbf{B} + \mathbf{A} \\ \mathbf{s}_i \geq 0 \\ \mathbf{1}_K^T\mathbf{s}_i = 1. \end{cases} \tag{6}$$

Inspired by the linear mixing model, the AE is augmented to construct an MDMN with a sparse regularizer for robust HAD. The anomaly component $\mathbf{A}$ can be culled out with the sparse regularizer, and the background component $\mathbf{B}$ can be represented by the combination of low-dimensional representation extracted by a union of AEs. Equation (6) can be rewritten as

$$\min_{\mathbf{\Theta},\mathbf{A}} \sum_{i=1}^{N} \Big[||\mathbf{b}_i - \left[\mathcal{AE}^{(1)}(\mathbf{b}_i), \mathcal{AE}^{(2)}(\mathbf{b}_i), \ldots, \mathcal{AE}^{(K)}(\mathbf{b}_i)\right]\mathbf{s}_i||_2^2 + \lambda_1 g_1(\mathbf{s}_i)\Big] + \lambda_2 g_2(\mathbf{A})$$
$$\text{s.t.} \begin{cases} \mathbf{X} = \mathbf{B} + \mathbf{A} \\ \mathbf{s}_i \geq 0 \\ \mathbf{1}_K^T\mathbf{s}_i = 1 \end{cases} \tag{7}$$

to implicitly model the spectral variability of background endmembers with a union of AEs, i.e., $\{\mathcal{AE}^{(i)}, i = 1, \ldots, K\}$, in which a single low-dimension manifold corresponding to a

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(a)

Algorithm: Alternating Optimization Algorithm of MDMN

Input: HSI $\mathbf{X} \in \mathbb{R}^{L\times N}$, the number of iterations M;

Output: Background component $\mathbf{B}$, anomaly component $\mathbf{A}$;

$\mathbf{A}^{(0)} = \text{zeros}(L \times N)$;

for $m = 1 : M$ **do**

%B-step: updating background component

$\mathbf{B}^{(m)} = \mathbf{X} - \mathbf{A}^{(m-1)}$;

Updating endmembers by Eq. (8) and (9);

Updating abundance by Eq. (10) and (11);

Updating parameters of MDMN by Eq. (19);

Updating $\mathbf{B}^{(m)}$ by Eq. (20);

%A-step: updating anomaly component

$\mathbf{X}_{\mathbf{B}}{}^{(m)} = \mathbf{X} - \mathbf{B}^{(m)}$;

Updating $\mathbf{A}^{(m)}$ by $\mathbf{A}^{(m)} = prox_{\lambda_2,l_{2,1}}(\mathbf{X}_{\mathbf{B}}{}^{(m)})$;

end for

$\mathbf{B} = \mathbf{B}^{(M)}$, $\mathbf{A} = \mathbf{A}^{(M)}$;

(b)

Fig. 3. (a) Schematic of the proposed MDMN and (b) alternating optimization algorithm of MDMN. The optimization process of MDMN consists of two substeps (B-step and A-step) for updating the background component with spectral mixing and endmember spectral variability, and the anomaly component, respectively, during each iteration. The network modules in MDMN produce outputs with clear interpretation, enabling in-depth analysis of the network's behavior during training and providing a comprehensive understanding of its functionality.

certain endmember can be characterized by a separated AE. Here, $\mathbf{s}_i$ is the fractional abundance vector at spectrum $\mathbf{b}_i$, g_1 is a regularizer which introduces sparsity in $\mathbf{s}_i$ to implicitly identify the most dominant endmembers in the given spectrum $\mathbf{b}_i$, $g_2 = \|\cdot\|_{2,1}$ is a structured sparsity regularizer to enforce the detection of specific anomalies, and λ_1 and λ_2 are the hyperparameters. More details about the proposed MDMN will be discussed in subsection (B), (C), and (D).

B. Model Architecture

In HSI processing, two assumptions motivate us to construct the MDMN to explore the intrinsic characterization for HSI background modeling and anomaly detection: 1) each category of background spectra is sampled from an embedded different low-dimensional nonlinear manifold and 2) anomalies tend to be small in size of proportion and low in probability of occurrence, and therefore exhibit spectrum-specific sparsity, i.e., only a few spectra are anomalous and the others are normal.

According to assumption (1), the embedded low-dimensional nonlinear manifold that characterizes the variability of a certain background category can be captured by a trained AE without any strong distribution assumptions. Thus, the MDMN, which regularizes a separate AE to characterize a type of background cluster and consequently represents the entire background with a union of AEs (i.e., the deep mixture module denoted as **MixNet**), is proposed to model the background as a mixture of low-dimensional manifold embedded in the original high-dimensional space.

The architecture of the proposed MDMN is illustrated in Fig. 2. The **MixNet** consists of a union of K AEs (i.e., $\{\mathcal{AE}^{(i)}, i = 1, \ldots, K\}$) with the same structure. For $\mathcal{AE}^{(k)}$, both the encoder $\mathbf{En}^{(k)}$ and the decoder $\mathbf{De}^{(k)}$ are neural networks as follows:

$$\mathbf{z}_i^{(k)} = \mathbf{En}^{(k)}\left(\mathbf{b}_i; \boldsymbol{\theta}_{en}^{(k)}\right) \tag{8}$$

$$\hat{\mathbf{e}}_i^{(k)} = \mathbf{De}^{(k)}\left(\mathbf{z}_i^{(k)}; \boldsymbol{\theta}_{de}^{(k)}\right) \tag{9}$$

where $\mathbf{b}_i \in \mathbb{R}^L, (i = 1, 2, \ldots, N)$ is sampled from the background component $\mathbf{B}$, $\hat{\mathbf{e}}_i^{(k)}$ is the kth endmember with respect to $\mathbf{b}_i$, $\boldsymbol{\theta}_{en}^{(k)}$ and $\boldsymbol{\theta}_{de}^{(k)}$ are the parameters of $\mathbf{En}^{(k)}$ and $\mathbf{De}^{(k)}$, respectively, and $\boldsymbol{\theta}_k = \{\boldsymbol{\theta}_{en}^{(k)}, \boldsymbol{\theta}_{de}^{(k)}\}$ is the parameter of $\mathcal{AE}^{(k)}$. Here, the physical interpretation of $\mathbf{z}_i^{(k)}$ can be a quantified value of the cause of the spectral variability at the location of the ith spectrum, such as the composition or morphology of the material, the size of an object, illumination, and atmospheric distortion.

To regularize an individual AE to primarily characterize a specific type of background land cover, an extra neural network termed soft-membership assignment network (**AssignNet**) is introduced, which adopts softmax as its last layer to guarantee ANC and ASC to predict the abundance for $\mathbf{x}_i$ based on K AEs' latent representation. Particularly, for $\mathbf{x}_i$, the concatenation of the latent representation vectors of K AEs, $\mathbf{z}_i^c$, is fed to the **AssignNet**

$$\mathbf{z}_i^c = \begin{bmatrix} \mathbf{z}_i^{(1)} \\ \mathbf{z}_i^{(2)} \\ \vdots \\ \mathbf{z}_i^{(K)} \end{bmatrix} \tag{10}$$

$$\mathbf{s}_i = \mathbf{AssignNet}\left(\mathbf{z}_i^c; \boldsymbol{\theta}_a\right) \tag{11}$$

where $\boldsymbol{\theta}_a$ is the parameter of the **AssignNet**, and $\mathbf{s}_i = [s_i^{(1)}, s_i^{(2)}, \ldots, s_i^{(K)}]^T$ is the abundance of $\mathbf{x}_i$ over all types of background endmembers. An algorithm flowchart for the forward propagation of MDMN is shown in Algorithm 1, which reveals the relationship among **MixNet**, **AssignNet**, and linear mixing model in detail.

Rather than separately modeling the background and then learning the abundance for background land covers, the proposed MDMN adopts a joint optimization paradigm that models the background representation as a union of low-dimensional manifolds using K AEs and performs abundance estimation simultaneously. The composite objective function of MDMN with the joint optimization paradigm is

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Algorithm 1 Forward Propagation Flowchart of MDMN

Input: Background component $\mathbf{B} \in \mathbb{R}^{L\times N}$, $batch_size$, the number of AEs K;
Output: Background component $\hat{\mathbf{B}}$;

for $i = 1 : \lceil \frac{N}{batch_size} \rceil$ **do**
  Sampling $\mathcal{B} \subseteq \mathbf{B}$, where $|\mathcal{B}| = batch_size$;
  for $j = 1 : batch_size$ **do**;
    $\mathbf{b}_j = \mathcal{B}[j]$
    for $k = 1 : K$ **do**; %MixNet
      $\mathbf{z}_j^{(k)} = \mathbf{En}(\mathbf{b}_j; \boldsymbol{\theta}_{en}^{(k)})$;
      $\hat{\mathbf{b}}_j^k = \mathbf{De}(\mathbf{z}_j^{(k)}; \boldsymbol{\theta}_{de}^{(k)})$
    end for
    $\mathbf{z}_i^c = [\mathbf{z}_i^{(1)\mathsf{T}}, \ldots, \mathbf{z}_i^{(K)\mathsf{T}}]^\mathsf{T}$;
    $\mathbf{s}_j = \mathbf{AssignNet}(\mathbf{z}_j^c; \boldsymbol{\theta}_a)$; %AssignNet
    $\hat{\mathbf{b}}_j = [\hat{\mathbf{b}}_j^{(1)}, \ldots, \hat{\mathbf{b}}_j^{(K)}]\mathbf{s}_j$; %Linear Mixing Model
    $\hat{\mathcal{B}}[j] = \hat{\mathbf{b}}_j$
  end for
  $\hat{\mathbf{B}} \supseteq \hat{\mathcal{B}}$
end for

formulated as

$$\min_{\boldsymbol{\Theta}} \sum_{i=1}^{N} \Big[||\mathbf{b}_i - [\mathcal{AE}^{(1)}(\mathbf{b}_i), \mathcal{AE}^{(2)}(\mathbf{b}_i), \ldots, \mathcal{AE}^{(K)}(\mathbf{b}_i)]\mathbf{s}_i||_F + \lambda_1 g_1(\mathbf{s}_i) \Big]$$
$$\text{s.t.} \quad \mathbf{X} = \mathbf{B} + \mathbf{A} \tag{12}$$

where $\boldsymbol{\Theta} = \{\boldsymbol{\theta}_k, \boldsymbol{\theta}_a | k = 1, \ldots, K\}$ refers to the parameters of MDMN, and $g_1(\cdot)$ aims to promote sparse abundance and identify the dominant endmembers in the given spectrum

$$g_1(\mathbf{s}_i) = -\sum_{k=1}^{K} s_i^{(k)} \log\left(s_i^{(k)}\right). \tag{13}$$

To avoid the trivial solution that the **AssignNet** predicts a constant abundance for different input spectra, i.e., a single low-dimensional representation extracted by a certain AE is select for representing all the categories of background spectra, an extra equal-usage regularizer, $eu(\cdot)$, is applied on a batch of input spectra

$$eu(\bar{\mathbf{s}}_{\mathcal{B}}) = \sum_{k=1}^{K} \bar{s}_{\mathcal{B}}^{(k)} \log\left(\bar{s}_{\mathcal{B}}^{(k)}\right) \tag{14}$$

where $\bar{\mathbf{s}}_{\mathcal{B}} = \sum_{i\in \text{ind}(\mathcal{B})}(\mathbf{s}_i/|\mathcal{B}|)$, $\mathcal{B} \subseteq \mathbf{B}$ is a randomly sampled minibatch, $|\mathcal{B}|$ is the cardinality of the finite set $\mathcal{B}$, $\text{ind}(\mathcal{B})$ is the set of index in $\mathcal{B}$, and $\bar{\mathbf{s}}_{\mathcal{B}}$ is the average abundance over the minibatch $\mathcal{B}$. Specifically, if (14) achieves a minimum value of $-\log(K)$, then $\bar{\mathbf{s}}_{\mathcal{B}} = (1/K)\mathbf{1}_K$, and each AE is selected with uniform probability.

Furthermore, according to observation (2), the observed HSI is sparsely corrupted by spectrum-specific anomalies. Based on the premise that all the training samples are normal, the existing unsupervised anomaly detection approaches such as [63], [64], and [48] explicitly or implicitly estimate the distribution of background and exhibit satisfactory performance on benchmark datasets. However, in HAD, a source of anomaly-free dataset for training is unavailable. Overfitting singular anomalies may prevent the model from correctly estimating the distribution of backgrounds and thus lead to a high miss detection ratio. Generally, model capacity reducing and early stopping can guide the model to focus on the majority (i.e., normal samples) of dataset and alleviate the negative effect of the corrupted dataset to some extent but trend to inaccurate distribution estimations and high false alarms. To better handle the corrupted dataset and accurately estimate background distribution, a structured sparsity regularizer $g_2(\cdot)$ is introduced into the MDMN

$$g_2(\mathbf{A}) = \|\mathbf{A}\|_{2,1} = \sum_{j=1}^{N} \sqrt{\sum_{i=1}^{L} \left([\mathbf{A}]_{i,j}\right)^2}. \tag{15}$$

The structured sparsity regularizer promotes the columns of $\mathbf{A}$ to be zero, which is based on the assumption that anomalies present spectrum-specific sparsity. With the introduced structured sparsity regularizer, MDMN can cull out anomalies from the original HSI during the training procedure, and various background classes can be accurately characterized as a mixture of embedded low-dimensional manifolds by the union of AEs.

Thus, the proposed MDMN can be mathematically formulated as an optimization problem

$$\min_{\boldsymbol{\Theta},\mathbf{A}} \sum_{\mathcal{B}\subseteq\mathbf{B}} \Bigg[\sum_{i\in\text{ind}(\mathcal{B})} \Big[||\mathbf{b}_i - \big[\mathcal{AE}^{(1)}(\mathbf{b}_i), \ldots, \mathcal{AE}^{(K)}(\mathbf{b}_i)\big]\mathbf{s}_i||_F + \lambda_1 g_1(\mathbf{s}_i) \Big] + \lambda_2 eu(\bar{\mathbf{s}}_{\mathcal{B}}) \Bigg] + \lambda_3 g_2(\mathbf{A})$$
$$\text{s.t.} \quad \mathbf{X} = \mathbf{B} + \mathbf{A} \tag{16}$$

where λ_1, λ_2, and λ_3 are the hyperparameters. The choice of the hyperparameters significantly affects the final background modeling quality and anomaly detection accuracy. Intuitively, the reconstruction error minimization is initially prioritized to encourage better learning of the manifolds for background spectra. Gradually, the minimization of $g_1(\cdot)$ and $eu(\cdot)$ should be prioritized to promote sparse abundances for the given spectra and equal usage of AEs, respectively. In this article, a dynamic adjustment scheme is adopted to adjust hyperparameters: comparatively small values are adopted to initialize λ_1 and λ_2 at the beginning of the training procedure, and they are adjusted in the tth epoch as the following exponential growth function:

$$\lambda_i = \lambda_{\text{iInit}} \gamma^t, \quad i \in \{1, 2\} \tag{17}$$

where λ_{iInit} is the initial value of λ_i, and γ is the growth rate that is set as 1.0005. Empirically, the dynamic adjustment scheme produces better results than static choices of λ_i.

C. Optimization Algorithm

In this section, a detailed description of the proposed MDMN for solving problem (16) is provided. Inspired by the alternating direction method of multipliers [75], an algorithm

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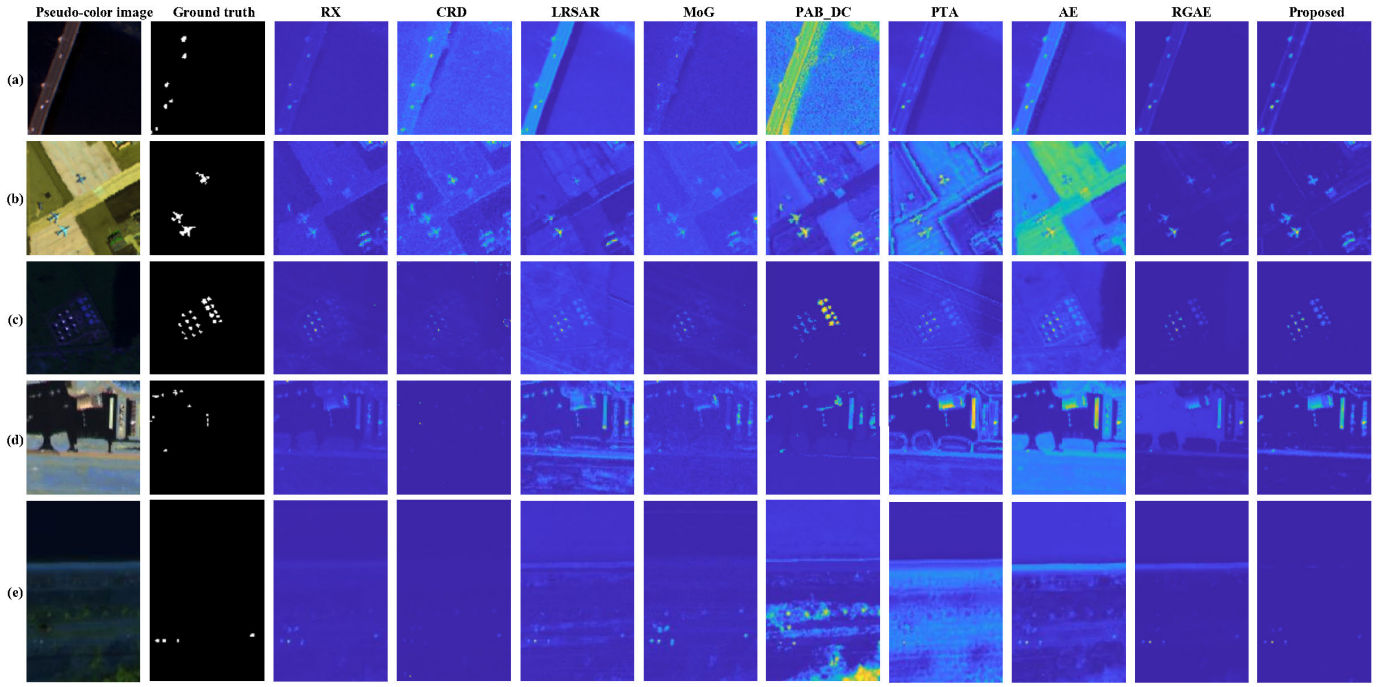


Fig. 4. Pseudocolor image, ground truth, and detection maps of the compared methods for (a) Pavia, (b) SanDiego, (c) Coast, (d) Gainesville, and (e) Honghu.

with simple computations easy to embed in neural networks is proposed to iteratively update each term in (16). The variable splitting technique is adopted to split the objective function in (16) into two parts: the former part depends on $\mathbf{B}$ and independent of $\mathbf{A}$, while the latter part, $\lambda_3 g_2(\mathbf{A})$, depends on $\mathbf{A}$ and independent of $\mathbf{B}$. Thus, the training procedure can be separately decomposed as the following two iterative steps and achieve the following steps for the mth stage of MDMN:

1) B-Step Updating: The background component $\mathbf{B}$ can be updated by solving the following optimization problem using backpropagation [76]:

$$\mathbf{B}^{(m)} = \mathbf{X} - \mathbf{A}^{(m-1)} \tag{18}$$

$$\boldsymbol{\Theta}^{(m)} = \arg\min_{\boldsymbol{\Theta}^{(m)}} \sum_{\mathcal{B}\subseteq\mathbf{B}^{(m)}} \left[\sum_{\mathbf{b}_i\in\mathcal{B}} \left[||\mathbf{b}_i - \left[\mathcal{AE}^{(1)}(\mathbf{b}_i), \ldots, \mathcal{AE}^{(K)}(\mathbf{b}_i)\right]\mathbf{s}_i||_F + \lambda_1 g_1(\mathbf{s}_i) \right] + \lambda_2 eu(\bar{\mathbf{s}}_{\mathcal{B}}) \right] \tag{19}$$

$$\mathbf{B}^{(m)} = \left[\mathbf{b}_1^{(m)}, \mathbf{b}_2^{(m)}, \ldots, \mathbf{b}_N^{(m)}\right] \tag{20}$$

where $\mathbf{b}_i^{(m)} = [\mathcal{AE}^{(1)}(\mathbf{b}_i; \boldsymbol{\theta}_1^{(m)}), \ldots, \mathcal{AE}^{(K)}(\mathbf{b}_i; \boldsymbol{\theta}_K^{(m)})]\mathbf{s}_i$, and $\mathbf{s}_i = \mathbf{AssignNet}(\mathbf{z}_i^c; \boldsymbol{\theta}_a^{(m)})$.

2) A-Step Updating: The anomaly component $\mathbf{A}$ can be updated by solving the following proximal problem [77]:

$$\mathbf{A}^{(m)} = \arg\min_{\mathbf{A}} \lambda_3 \|\mathbf{A}\|_{2,1} + \frac{1}{2}\|\mathbf{A} - (\mathbf{X} - \mathbf{B}^{(m)})\|_F^2. \tag{21}$$

The $l_{2,1}$-norm can be optimized efficiently through the use of the proximal method, which can be written as the following blockwise soft-thresholding function:

$$\mathbf{A}^{(m)} = \left[\text{prox}_{\lambda_3, l_{2,1}}(\mathbf{X}_{\mathbf{B}}^{(m)})\right]_{:,i} = \begin{cases} 0, & \text{if } \left\| \left[\mathbf{X}_{\mathbf{B}}^{(m)}\right]_{:,i} \right\|_2 \le \lambda_3 \\ \dfrac{\left\| \left[\mathbf{X}_{\mathbf{B}}^{(m)}\right]_{:,i} \right\|_2 - \lambda_3}{\left\| \left[\mathbf{X}_{\mathbf{B}}^{(m)}\right]_{:,i} \right\|_2} \left[\mathbf{X}_{\mathbf{B}}^{(m)}\right]_{:,i}, & \text{otherwise} \end{cases} \tag{22}$$

where $\mathbf{X}_{\mathbf{B}}^{(m)} = \mathbf{X} - \mathbf{B}^{(m)}$. $[\text{prox}_{\lambda_3, l_{2,1}}(\mathbf{X}_{\mathbf{B}}^{(m)})]_{:,i}$ and $[\mathbf{X}_{\mathbf{B}}^{(m)}]_{:,i}$ are the ith column vector of matrix $\text{prox}_{\lambda_3, l_{2,1}}(\mathbf{X}_{\mathbf{B}}^{(m)})$ and $\mathbf{X}_{\mathbf{B}}^{(m)}$, respectively. This optimization encourages the columns of $\mathbf{A}$ to be zero, i.e., the structured sparsity, which is the spectrum-specific sparsity prior. The optimization of MDMN is summarized as Fig. 3.

Once $\mathbf{X}$ is decomposed into background component $\mathbf{B}$ and anomaly component $\mathbf{A}$, the anomaly score of $\mathbf{x}_i$ can be determined as $||[\mathbf{A}]_{:,i}||_2$, where $[\mathbf{A}]_{:,i}$ is the ith column of matrix $\mathbf{A}$.

D. Theoretical Explanation

Motivated by ensemble learning [78], [79], we can prove the superiority of combining multiple AEs in our proposed MDMN by considering the following three scenarios: 1) individual AEs; 2) multiple AEs combined based on a simple average strategy; and 3) the proposed MDMN. First, we analyze the performance of scenarios 1) and 2) and demonstrate that the reconstruction error of scenario 2) is smaller than that

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TABLE I
NETWORK CONFIGURATION IN EACH LAYER OF THE PROPOSED MDMN

Network			Input Ch.	Output Ch.	Filter Size	Stride
the kth Autoencoder of **MixNet**	Encoder	Conv1	L	100	1×1	1
		Conv2	100	50	1×1	1
		Conv3	50	10	1×1	1
	Decoder	Conv4	10	50	1×1	1
		Conv5	50	100	1×1	1
		Conv6	100	L	1×1	1
AssignNet		Conv7	$10\times K$	100	1×1	1
		Conv8	100	50	1×1	1
		Conv9	50	K	1×1	1

TABLE II
QUANTITATIVE RESULTS FOR ABLATION STUDY

Configuration	K	$g_1(\cdot)$	$eu(\cdot)$	$g_2(\cdot)$	Average AUC of (TPR, FPR)	Average AUC of (FPR, τ)
(A)	1	✗	✗	✗	0.8522	0.1848
(B)	5	✗	✗	✗	0.8469	0.1150
(C)	1	✗	✗	✓	0.8391	0.1024
(D)	5	✗	✗	✓	0.9187	0.0848
(E)	5	✗	✓	✗	0.8413	0.1143
(F)	5	✓	✗	✗	0.8443	0.1161
(G)	5	✓	✓	✗	0.9354	0.0803
(H)	5	✓	✓	✓	**0.9892**	**0.0128**

of scenario 1). Then we analyze the adaptive weighted average strategy adopted in the proposed MDMN, which can be proven superior to scenarios 1) and 2).

In the proposed MDMN, a set of K individual AEs $\{\mathcal{AE}^{(k)}|k = 1, \ldots, K\}$ is given, in which each AE approximates the underlying true hypothesis f, and the output of $\mathcal{AE}^{(k)}$ for background spectrum $\mathbf{x} \sim p(\mathbf{x})$ is $\mathcal{AE}^{(k)}(\mathbf{x}) \in \mathbb{R}^L$. We first consider scenario 1). The output of the kth AE can be formulated as the true hypothesis f plus an error item $\epsilon_k(\mathbf{x})$, i.e.,

$$\mathcal{AE}^{(k)}(\mathbf{x}) = f(\mathbf{x}) + \epsilon_k(\mathbf{x}), \quad (k = 1, \ldots, K). \tag{23}$$

And the mean squared error of an individual AE ($\mathcal{AE}^{(k)}$) is

$$\text{mse}_{(a)} = E_{\mathbf{x}\sim p(\mathbf{x})}\left[\epsilon_k(\mathbf{x})^2\right] = \int \epsilon_k(\mathbf{x})^2 p(\mathbf{x}) d\mathbf{x}. \tag{24}$$

Therefore, the average error made by the individual AEs $\{\mathcal{AE}^{(k)}|k = 1, \ldots, K\}$ is

$$\overline{\text{mse}}_{(a)} = \frac{1}{K}\sum_{k=1}^{K}\int \epsilon_k(\mathbf{x})^2 p(\mathbf{x}) d\mathbf{x}. \tag{25}$$

In scenario 2), the simple average strategy obtains the output by combining the output of individual AEs directly as follows:

$$H(\mathbf{x}) = \frac{1}{K}\sum_{k=1}^{K}\mathcal{AE}^{(k)}(\mathbf{x}). \tag{26}$$

The expectation of the mean squared error of (26) is

$$\begin{aligned}\text{mse}_{(b)} &= \int\left(\frac{1}{K}\sum_{k=1}^{K}\mathcal{AE}^{(k)}(\mathbf{x}) - f(\mathbf{x})\right)^2 p(\mathbf{x})d\mathbf{x} \\ &= \frac{1}{K^2}\int\left(\sum_{k=1}^{K}\epsilon_k(\mathbf{x})\right)^2 p(\mathbf{x})d\mathbf{x}.\end{aligned} \tag{27}$$

Based on the Cauchy–Schwarz inequality [80], (27) can be formulated as

$$\text{mse}_{(b)} \le \frac{1}{K}\sum_{k=1}^{K}\int \epsilon_k(\mathbf{x})^2 p(\mathbf{x})d\mathbf{x} = \overline{\text{mse}}_{(a)} \tag{28}$$

which implies that the expectation of the mean squared error of scenario 2) is no larger than the averaged error of scenario 1). Moreover, if we assume that the error items $\{\epsilon_k(\mathbf{x})|k = 1, \ldots, K\}$ are mutually independent with zero mean, i.e.,

$$\int \epsilon_k(\mathbf{x})p(\mathbf{x})d\mathbf{x} = 0 \quad \text{and} \int \epsilon_k(\mathbf{x})\epsilon_j(\mathbf{x})p(\mathbf{x})d\mathbf{x} = 0, \ (k \ne j). \tag{29}$$

Equation (27) can be derived as

$$\text{mse}_{(b)} = \frac{1}{K^2}\int\sum_{k=1}^{K}\epsilon_k(\mathbf{x})^2 p(\mathbf{x})d\mathbf{x} = \frac{1}{K}\overline{\text{mse}}_{(a)} \tag{30}$$

which implies that in scenario 2), by simply combining multiple AEs on average, we can reduce the mean square error of scenario 1) by a factor of K.

Then, based on the above analysis, we analyze scenario 3), which obtains the combined output by averaging the outputs of multiple AEs with adaptive weights generated by the **AssignNet**

$$\text{MDMN}(\mathbf{x}) = \sum_{k=1}^{K} s^{(k)}\mathcal{AE}^{(k)}(\mathbf{x}) \tag{31}$$

where $s^{(k)}$ is the weight (also refers to the abundance) for $\mathcal{AE}^{(k)}$, and the weight vector $\mathbf{s} = [s^{(1)}, \ldots, s^{(k)}]^T \in \mathbb{R}^{K\times 1}$ assumes to be constrained by nonnegative constraint and sum-to-one constraint in (3). Similar to (27), the expectation of the mean squared error of scenario 3) [i.e., (31)] is

$$\begin{aligned}\text{mse}_{(c)} &= \int\left(\sum_{k=1}^{K}s^{(k)}\mathcal{AE}^{(k)}(\mathbf{x}) - f(\mathbf{x})\right)^2 p(\mathbf{x})d\mathbf{x} \\ &= \sum_{k=1}^{K}\sum_{j=1}^{K}s^{(k)}s^{(j)}\int\epsilon_k(\mathbf{x})\epsilon_j(\mathbf{x})p(\mathbf{x})d\mathbf{x}.\end{aligned} \tag{32}$$

According to (29), (32) can be formulated as

$$\begin{aligned}\text{mse}_{(c)} &= \sum_{k=1}^{K}(s^{(k)})^2\int(\epsilon_k(\mathbf{x}))^2 p(\mathbf{x})d\mathbf{x} \\ &\le \sum_{k=1}^{K}s^{(k)}\int(\epsilon_k(\mathbf{x}))^2 p(\mathbf{x})d\mathbf{x} = \sum_{k=1}^{K}s^{(k)}\text{mse}_{(a)}\end{aligned} \tag{33}$$

which implies that the expectation of the mean squared error of scenario 3) is no larger than the weighted averaged error of scenario 1).

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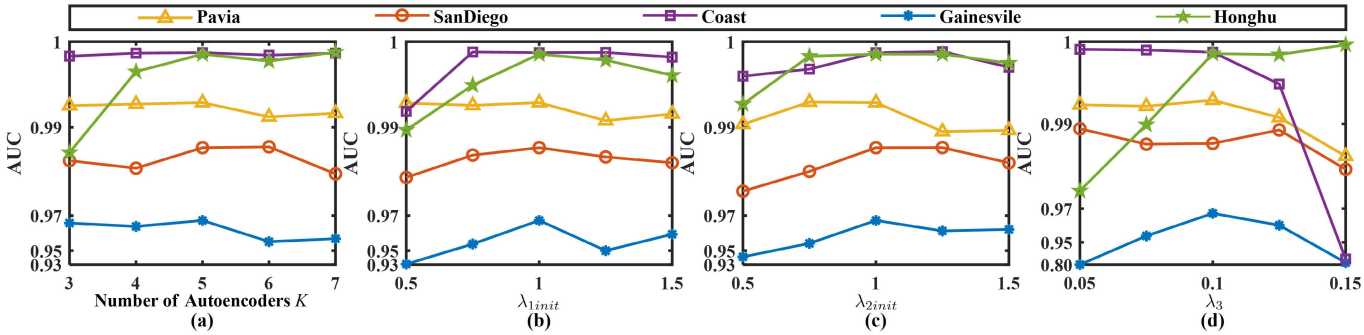


Fig. 5. Parameter sensitivity analysis of the influence on different datasets. (a) Number of AEs K, (b) $\lambda_{1\text{init}}$, (c) $\lambda_{2\text{init}}$, and (d) λ_3.

Furthermore, we assume that in scenario 3), in the ideal case, K AEs and the **AssignNet** converge to the optimal solution at the end of training. Therefore, the optimal weight output by the **AssignNet** is equivalent to the solution solved by minimizing (33), i.e.,

$$\mathbf{AssignNet}\left(\mathbf{z}^c; \hat{\boldsymbol{\theta}}_a\right) = \arg\min_{\mathbf{s}} \text{mse}_{(c)} \tag{34}$$

where $\mathbf{z}^c$ is defined in Section III-B, and $\hat{\boldsymbol{\theta}}_a$ is the optimal parameter of the **AssignNet**. Thus, we use the solution of the right-hand side of (34) to further analyze the performance of scenario 3). Here, the Lagrange multiplier method is applied to solve the optimization problem

$$\min_{\mathbf{s}} \mathbf{s}^T \mathbf{C}\mathbf{s} - \lambda(\mathbf{1}^T\mathbf{s} - 1) \tag{35}$$

where λ is the Lagrange multiplier, $\mathbf{C} = \text{diag}(C_1, \ldots, C_K)$ is a diagonal matrix, and $C_k = \int \epsilon_k(\mathbf{x})^2 p(x) dx$. And the solution is

$$s^{(k)} = C_k^{-1}\left(\sum_{j=1}^{K} C_j^{-1}\right)^{-1}. \tag{36}$$

Combining (33) and (36), the optimal mean square error expectation of scenario 3) is given by

$$\text{mse}_{(c)} = \sum_{k=1}^{K} (s^{(k)})^2 C_k = \left(\sum_{j=1}^{K} C_j^{-1}\right)^{-1}. \tag{37}$$

Thus, comparing (30) and (37), we have

$$\text{mse}_{(b)} = \frac{1}{K^2}\sum_{k=1}^{K} C_k \geq \left(\sum_{k=1}^{K} C_k^{-1}\right)^{-1} = \text{mse}_{(c)} \tag{38}$$

which can be proven using the Cauchy–Schwarz inequality. Thus

$$\overline{\text{mse}}_{(a)} \geq \text{mse}_{(b)} \geq \text{mse}_{(c)}. \tag{39}$$

Therefore, in the mean square error sense, the proposed MDMN [i.e., (31)], which combines multiple AEs based on the adaptive weighted average strategy, provides the best approximation of f comparing to the single AE in scenario 1) and the multiple AEs combined with the simple average strategy in scenario 2).

IV. Experimental Results and Discussion

A. Datasets

The proposed MDMN is assessed on four benchmark hyperspectral datasets with various scenarios of land covers.

1) Pavia Dataset: It was obtained by Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor over Pavia city. It consists of 100 × 100 pixels and 102 spectral bands covering the wavelength range from 430 to 860 nm, where the vehicles of 63 pixels are considered to be anomalies. The pseudocolor image and ground truth are presented in Fig. 4(a).

2) SanDiego Dataset: It was acquired by the Airborne Visible/ Infrared Imaging Spectrometer (AVIRIS) sensor over the area of San Diego, which consists of 100 × 100 pixels and 189 spectral bands ranging from 400 to 2500 nm. Three aircraft with structural information are anomalies. The pseudocolor image and ground truth are presented in Fig. 4(b).

3) Coast Dataset: Coast dataset captured by AVIRIS sensor was obtained in the Texas Coast. It contains 100 × 100 pixels, each with 207 channels in a wavelength ranging from 450 to 1350 nm. Buildings of different scales are regarded as anomalies. The pseudocolor image and ground truth are presented in Fig. 4(c).

4) Gainesville Dataset: It was also acquired by the AVIRIS sensor over the area of Gainesville city. Unsuitable bands in the original images have been removed. There are 100 × 100 pixels, with a 3.5-m spatial resolution and 191 spectral bands varying from 430 to 860 nm. A series of ships, which consists of 52 pixels in total, are considered to be anomalies. The pseudocolor image and ground truth are presented in Fig. 4(d).

5) Honghu Dataset: The Honghu dataset [81] was acquired by the RSIDEA Group at Wuhan University, Wuhan, China, using a NanoHyperspec sensor operating in the visible and near-infrared spectrum, mounted on an unmanned aerial vehicle. This dataset encompasses a spectral range spanning from 400 to 1000 nm, encompassing a total of 135 spectral bands. The dimensions of the image scene measure 105 × 168 pixels, with anomalies accounting for approximately 2.04% of the image area.

B. Experimental Setup

1) Comparison Methods: To reveal the performance of the proposed MDMN, eight classical and state-of-the-art methods,

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including RX [23], CRD [34], LRASR [41], MoG [28], PAB_dc [72], PTA [42], AE [82], and RGAE [53], are selected for comparison. All the methods are implemented according to their articles and open-source codes.

2) Performance Evaluation Metrics: To quantitatively evaluate the superiority and effectiveness of the proposed MDMN, the receiver operating characteristic (ROC) curve and the area under the ROC curve (AUC) are adopted as performance evaluation metrics. ROC and AUC have been widely used in the field of signal processing and machine learning. The ROC curve is generated by three parameters, i.e., true positive rate (TPR), false positive rate (FPR), and threshold (τ). TPR and FPR are the detection rate and false alarm rate under a certain threshold, τ, respectively. The ROC curve of (TPR, FPR) allows users to describe the effectiveness of a method as a function of TPR versus FPR through the implicit parameter τ. In real applications, visually comparing the ROC curves of different methods cannot provide quantitative analysis. To address this issue, the area under the ROC curve (AUC) of (TPR, FPR) is adopted as an alternative metric to quantitatively evaluate the detection performance. The larger the AUC of (TPR, FPR), the better the detection performance. Besides, the AUC of (FPR, τ) is introduced to provide a quantitative assessment of the false alarm. The smaller the AUC of (FPR, τ), the lower the false alarm rate. Moreover, box–whisker plot, which provides the variation in samples of a statistical population without any assumption of the underlying statistical distribution, is adopted to assess the separability between anomalies and background. The spacings between different parts of a box assess the degree of dispersion and skewness of the samples. More details about box–whisker plot can be found in [83].

3) Implementation Details: The proposed MDMN is composed of a **MixNet** with a union of K AEs and an **AssignNet**. Rhe specific network configurations of it are detailed in Fig. 2 and Table I. The number of AEs, K, is empirically set to 5. And the latent dimension of K AEs is set to 10. The MDMN is trained by the stochastic gradient descent (SGD) optimizer with the learning rate set to 5×10^{-4} and the batch size set to 1000. The initial value of the hyperparameters λ_1 and λ_2 are 1 and 0.9, respectively, and λ_3 is empirically set as 0.1. The MDMN is implemented by PyTorch on an NVIDIA 2080 Ti GPU with 11-GB memory.

C. Parameter Sensitivity Analysis

The proposed MDMN involves a parameter K which indicates the number of AEs and three hyperparameters in (16): λ_1, λ_2, and λ_3. As discussed in Section III, hyperparameters λ_1 and λ_2 are adjusted in a dynamic manner during the training procedure. Thus, the sensitivity of the initial values of λ_1 and λ_2, i.e., $\lambda_{1\text{init}}$ and $\lambda_{2\text{init}}$, is analyzed. Besides, when a certain parameter is being analyzed, the other parameters are fixed as their optimal values of $K = 5$, $\lambda_{1\text{init}} = 1$, $\lambda_{2\text{init}} = 1$, and $\lambda_3 = 0.1$. As can be seen in Fig. 5, the horizontal axis stands for the values of the parameters, and the vertical axis refers to the corresponding AUCs of (TPR, FPR).

1) Number of AEs K: For the number of AEs K, the value is ranging from 3 to 7 for all the datasets to perform parameter sensitivity analysis for the proposed MDMN. As shown in Fig. 5(a), for a smaller value of K, the proposed MDMN is unable to obtain a discriminative representation for various background land covers and comprehensively characterize the background distribution, while a larger value of K may introduce unnecessary redundant information and noise from the raw datasets, and thus degrades the detection performance. With a proper K value of 5, the proposed MDMN can capture a series of low-dimensional manifolds that characterize the data structure of various background by the K trained AEs and promote the detection performance.

2) Hyperparameters λ_{1init}, λ_{2init}, and λ_3: A larger $\lambda_{1\text{init}}$, $\lambda_{2\text{init}}$, and λ_3 would lead to insufficient training of K AEs and ineffective learning of K low-dimensional manifolds and thus weaken the detection performance. However, a smaller value of $\lambda_{1\text{init}}$ cannot identify one dominant assignment for each spectrum ultimately, and a smaller $\lambda_{2\text{init}}$ cannot effectively avoid the trivial solution, in which the **AssignNet** predicts a constant abundance for various input spectra. In addition, a smaller value of λ_3 would make the model overfit singular anomalies and prevent the model from correctly estimating background distribution, such that the detection results would exhibit a high miss detection ratio. Whit a proper value of $\lambda_{1\text{init}}$, $\lambda_{2\text{init}}$, and λ_3, the proposed MDMN can avoid overfitting singular anomalies and converge to a better state that the distribution of background can be well-characterized. As illustrated in Fig. 5(b)–(d), $\lambda_{1\text{init}} = 1$, $\lambda_{2\text{init}} = 1$, and $\lambda_3 = 0.1$ are selected as an optimal setting.

D. Ablation Study

In this section, an ablation study is conducted on the four datasets to evaluate the contribution of different components in the proposed MDMN. In detail, eight scenarios are designed as follows.

1) An individual AE described in Section III-D scenario 1) detects anomalies by calculating reconstruction errors between the original spectra and the reconstructed ones.
2) K AEs combined based on a simple average strategy described in Section III-D scenario 2) detect anomalies by calculating reconstruction errors.
3) The proposed MDMN with only one AE to characterize background distribution, i.e., scenario (A) regularized with g_2.
4) K AEs combined based on a simple average strategy regularized with g_2, i.e., scenario (B) regularized with g_2.
5) The proposed MDMN eliminates $g_1(\cdot)$ and $g_2(\cdot)$.
6) The proposed MDMN eliminates $eu(\cdot)$ and $g_2(\cdot)$.
7) The proposed MDMN eliminates $g_2(\cdot)$, which is scenario 3) in Section III-D.
8) The proposed MDMN.

The average evaluation results on all the four datasets are summarized in Table II.

As summarized in Table II, the proposed MDMN achieves a 0.9892 AUC of (TPR, FPR) on average and a 0.0128 AUC of (FPR, τ) on average. For the individual AE as can be seen in Table II(A), determining anomalies by calculating

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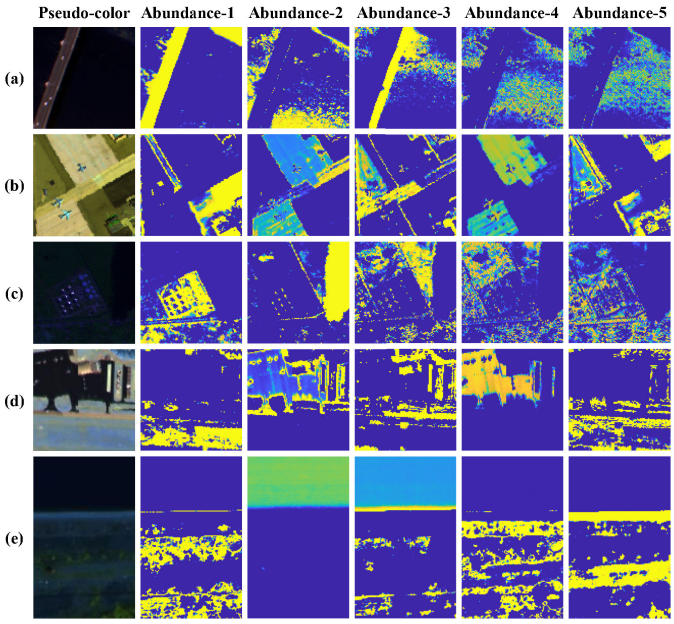


Fig. 6. Abundance maps of unmixing results for (a) Pavia, (b) SanDiego, (c) Coast, (d) Gainesville, and (e) Honghu.

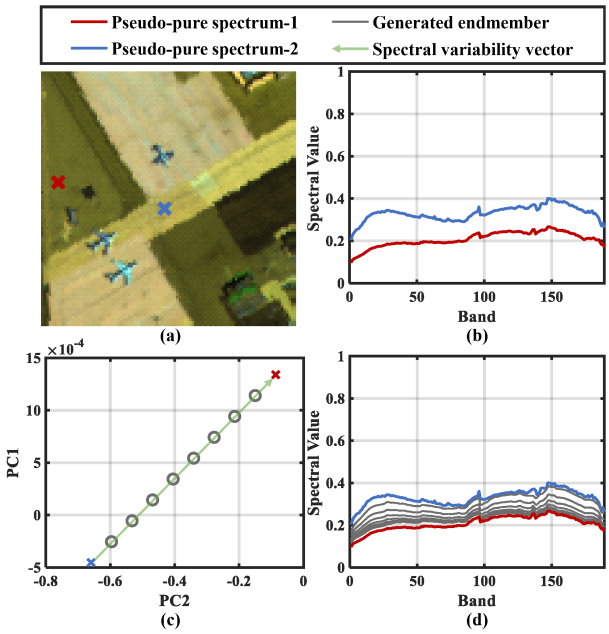


Fig. 7. Interpretation of the latent space for spectral variability for SanDiego dataset. (a) False color of the dataset and two spectra positions. (b) Two spectral curves. (c) Latent vectors in 2-D space projected by PCA. (d) Generated endmembers.

reconstruction error of an AE delivers a 0.8522 AUC of (TPR, FPR) on average and a 0.1848 AUC of (FPR, τ) on average, which indicates that the original AE is insufficient to characterize the complex background distribution and cull out the anomalies. Compared with (A), due to the better capability of (B) in terms of spectra reconstruction as discussed in Section III-D, it carries the risk of overfitting anomalies, such that it leads to a weaker performance than (A) as shown in Table II(B). Scenarios (C) and (D) are $g_2(\cdot)$ regularized versions of (A) and (B), respectively. Although the average AUC of (TPR, FPR) of (C) is slightly smaller than that of (A) due to the weak performance of an individual AE, scenario (C) acquires a lower false alarm than that of (A), which indicates that $g_2(\cdot)$ plays an important role in eliminating false alarms. Analogously, for scenario (D) shown in Table II(D), benefiting from the better capability of spectra reconstruction in (B) and $g_2(\cdot)$ regularization, the AUC of (TPR, FPR) is significantly increased to 0.9187, and the AUC of (FPR, τ) is reduced to 0.0848. By removing $g_2(\cdot)$ from the proposed MDMN as shown in Table II(G), the AUC of (TPR, FPR) is decreased to 0.9354, and the AUC of (FPR, τ) is increased to 0.0803, which also verifies the important role played by $g_2(\cdot)$ in background suppression and anomaly extraction. Besides, comparing (G) with (D), we find that **MixNet** in MDMN brings consistent improvement in revealing complicated background structures in HSIs, which is consistent with the theoretical explanation in Section III-D. Eliminating $g_1(\cdot)$ from scenario (G) will degenerate to a trivial solution where all the K AEs learn to reconstruct all the spectra without distinction, which is equivalent to scenario (B), such that the evaluation results of Table II(B) and (E) are almost equal. And discarding $eu(\cdot)$ from scenario (G) will lead to another trivial solution that the **AssignNet** predicts a constant one-hot abundance vector for different input spectra, i.e., only an individual AE is selected for representing all the categories of background spectra, while the others are not converged, which is an approximation to scenario (A). This is evidenced by the similar evaluation results of Table II(A) and (F). Overall, scenarios (E) and (F) verifies the necessity of introducing regularizers $g_1(\cdot)$ and $eu(\cdot)$.

E. Model Verification

The interpretability of the proposed MDMN, which facilitates easy analysis of the working mechanism inside the model, is verified in this section.

Fig. 6 visualizes the fractional abundance estimation for the four datasets. Due to the lack of background prior information, rather than specifying exactly the number of background endmembers, we empirically set it to 5. It can be seen that our proposed method can obtain meaningful abundance results for the materials. The bright pixels in one abundance map are very likely to be dim in the remaining abundance maps; thus, the extracted abundance maps are almost orthogonal to each other. This should be attributed to the proper guidance of the proposed **MixNet** that reveals the prior structures of background with spectral mixing and spectral variability.

Fig. 7 presents a possible physical interpretation of the extracted low-dimensional manifold of an endmember. We manually extract two pseudopure soil spectra under different composition and moisture content [84] conditions from the scene and labeled their location in Fig. 7(a). The corresponding spectral curves are shown in Fig. 7(b). Then, the encoder corresponding to the soil endmember identifies the low-dimensional representations of each spectrum, which are projected to 2-D space by PCA [85] for visualization, as shown in Fig. 7(c). We establish a "spectral variability vector," which represents the spectral variability of the soil endmember caused by different compositions and moisture content. The "spectral variability vector" is directed from the lowest reflectance spectrum (red) to the highest reflectance

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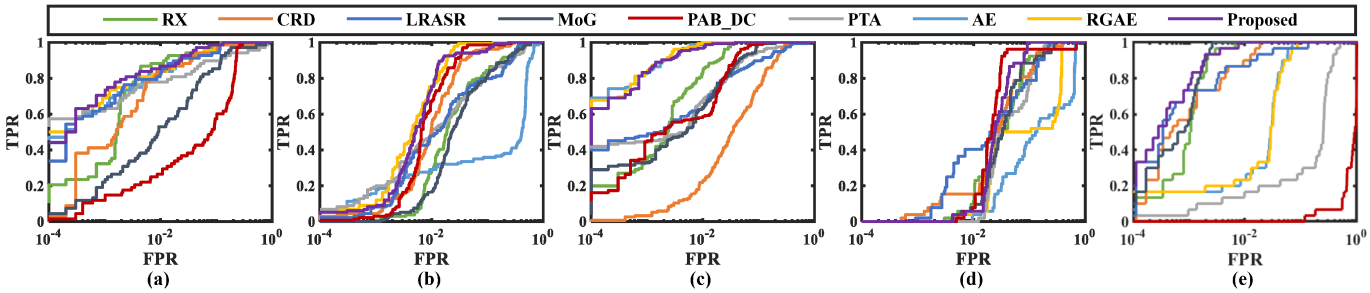


Fig. 8. ROC curves for five hyperspectral dataset. (a) Pavia. (b) SanDiego. (c) Coast. (d) Gainesville. and (e) Honghu.

TABLE III
AUC SCORES OF (TPR, FPR) FOR THE ALGORITHMS ON THE EXPERIMENTAL DATASETS

Image	RX	CRD	LRASR	MoG	PAB_DC	PTA	AE	RGAE	Proposed
Pavia	0.9887	0.9815	0.9892	0.9520	0.9000	0.9516	0.9897	0.9914	**0.9934**
SanDiego	0.9403	0.9746	0.9343	0.9304	0.9861	0.9316	0.6909	**0.9920**	0.9868
Coast	0.9946	0.9304	0.9637	0.9841	0.9937	0.9770	0.9903	0.9990	**0.9990**
Gainesvile	0.9512	0.9397	0.9476	0.9478	0.9563	0.9385	0.7201	0.8192	**0.9681**
Honghu	0.9884	0.9835	0.9893	0.9888	0.4560	0.7558	0.9655	0.9662	**0.9988**
Average	0.9726	0.9619	0.9702	0.9606	0.8584	0.9109	0.8712	0.9535	**0.9892**

TABLE IV
AUC SCORES OF (FPR, τ) FOR THE ALGORITHMS ON THE EXPERIMENTAL DATASETS

Image	RX	CRD	LRASR	MoG	PAB_DC	PTA	AE	RGAE	Proposed
Pavia	0.0268	0.1560	0.1075	0.0577	0.3425	0.0524	0.0178	0.0189	**0.0080**
SanDiego	0.0599	0.1087	0.0752	0.1139	0.1193	0.1882	0.3917	0.0392	**0.0169**
Coast	0.0145	0.0155	0.0871	0.0193	0.0067	0.0695	0.0759	0.0433	**0.0038**
Gainesvile	0.0360	**0.0033**	0.0882	0.0632	0.0383	0.0979	0.2016	0.0084	0.0325
Honghu	0.0162	0.0050	0.0407	0.0260	0.1001	0.1407	0.1139	0.0090	**0.0032**
Average	0.0306	0.0577	0.797	0.0560	0.1213	0.1097	0.1601	0.0237	**0.0128**

spectrum (blue) between the two spectra in the latent space. Furthermore, an interpolation on this vector is made and fed to the decoder of soil endmember to generate some new endmembers under different levels of composition and moisture content, which may not exist in the scene, as shown in Fig. 7(d). The aforementioned example reveals that the physical interpretation of low-dimensional representation produced by an AE is a quantified value of the cause of endmember spectral variability, which verifies the reasonability of our method and embodies the peculiarity of our proposal.

F. Detection Performance

The detection results corresponding to various datasets are shown in Fig. 4. Tables III and IV and Fig. 8 indicate the AUCs of (TPR, FPR), the AUCs of (FPR, τ), and the ROC curves, respectively.

For the Pavia dataset, as shown in Fig. 4(a), both the proposed MDMN and RGAE can well detect the vehicles on the bridge with high intensity and relatively low background interference. The RX can also detect the vehicles but the separation between background and anomalies is not evident. The CRD, LRASR, PTA, and AE can detect the vehicles while they yield a high false alarm rate. The MoG and PAB_dc fail to identify the position of the vehicles. Tables III and IV indicate the AUCs of (TPR, FPR) and AUCs of (FPR, τ) by different methods for the Pavia dataset. As can be seen, the proposed MDMN outperforms other methods in terms of detection performance and background suppression.

For the SanDiego dataset, as shown in Fig. 4(b), both the proposed MDMN and RGAE can identify the position of the aircraft in the airstrip and suppress the background interference. However, for the RGAE, the edge details of some aircraft are not preserved. The LRASR, PAB_dc, and PTA can well highlight the aircraft but with a high false alarm rate. Besides, the detection results of RX, CRD, MoG, and AE methods are polluted to some degree by the noise due to the interference of the background. As shown in Tables III and IV, the proposed MDMN guarantees the highest value of the AUC of (TPR, FPR) and the lowest value of the AUC of (FPR, τ).

For the Coast dataset, as shown in Fig. 4(c), both the proposed MDMN and RGAE can detect all the buildings of different scales with high intensity and a low false alarm. The PAB_dc can well suppress the background interference but fails to preserve some small buildings. The detection results of LRASR and PTA are contaminated by stripe noise and Gaussian noise due to the failure of background suppression. The RX, CRD, and MoG cannot identify the locations of the buildings with large scales. According to the ROC curves and the AUCs in Fig. 8 and Tables III and IV, the proposed MDMN exhibits the best performance on the Coast dataset.

Fig. 4(d) reveals the detection results on the Gainesville dataset. LRASR, PTA, and AE cannot well suppress the background interference. The results of RX, LRASR, and MoG are contaminated by noise, and the ships are more likely to be confused by the background. The detection results of CRD and RGAE lose most of the anomalies visually. PAB_dc and the proposed MDMN can highlight the ships with satisfactory background suppression as can be seen in Tables III and IV.

Fig. 4(e) visualizes the detection results of the Honghu dataset obtained from all the comparison methods. PAB_dc and PTA failed to modeling background distribution and lost most of the anomalies. The visual results of RX, CRD, and

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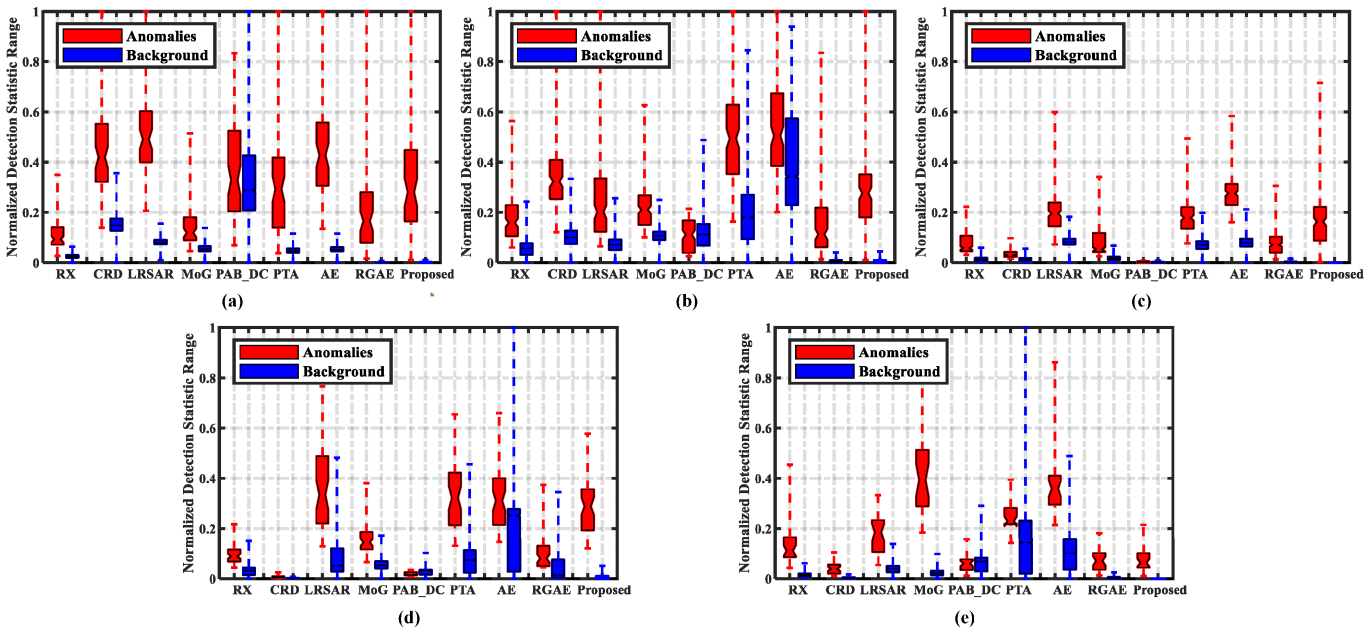


Fig. 9. Separability maps of the nine detectors for the datasets. (a) Pavia. (b) SanDiego. (c) Coast. (d) Gainesville. and (e) Honghu.

LRSAR are similar, while the separation between background and anomalies is not distinct. Both MDMN and RGAE can highlight the anomalies and suppress the background with high intensity and low false alarm, while MDMN provides the highest AUC of (TPR, FPR) and a lowest AUC of (FPR, τ) as shown Tables III and IV.

To further assess the separability between anomalies and background of the aforementioned methods, the box–whisker plots of the proposed MDMN and the comparison methods are plotted in Fig. 9. With the pixel values in a detection result and the corresponding ground truth, two boxes are plotted to enclose the main parts of the anomalous pixels and background pixels, respectively, excluding the biggest 10% and the smallest 10%. The anomaly and background columns for each method are plotted as red boxes and blue boxes, respectively. The position of the boxes reveals the overall tendency and compactness of the distribution of the pixels in the detection results. The separability of anomalies and background is evaluated by the relative position between the red box and the blue box. For the Pavia dataset as shown in Fig. 9 (a), the red box is overlapped with the blue box for PAB_dc and PTA. RX, CRD, and MoG cannot ensure the gap between the red box and the blue box. The gaps between the two boxes for LRASR, AE, RGAE, and MDMN are evident, and the separability of the proposed MDMN is more distinguishing. For the SanDiego dataset shown in Fig. 9(b), only the proposed MDMN can ensure evident gap between the anomaly box and the background box and successively distinguish anomalies from the suppressed background. As for the Coast dataset, Fig. 9(c) shows that all the methods, except CRD, PAB_dc, and PTA, have an obvious separation between the anomaly box and the background box. And MDMN guarantees a result with more effective background suppression in comparison to others. For the Gainesville dataset shown in Fig. 9(d), the boxes are overlapped for CRD, PAB_dc, and PTA. The gaps between the two boxes for RX, LRASR, RGAE, MoG, and MDMN are notable, and background suppression of MDMN is more satisfactory. Similarly, for the Honghu dataset, both MDMN and RGAE can better suppress the background interference and effectively separate the anomalies from backgrounds compared with other methods, as shown in Fig. 9(e).

TABLE V
COMPUTING TIME (SECONDS) OF DIFFERENT METHODS

Image	RX	CRD	LRASR	MoG	PAB_DC	PTA	AE	RGAE	Proposed
Pavia	**0.22**	68.22	41.94	2.35	360.57	88.07	35.57	51.65	212.08
SanDiego	**0.25**	67.31	30.44	2.21	366.71	89.31	33.77	57.03	206.33
Coast	**0.21**	70.18	33.76	2.76	370.53	85.03	33.36	54.69	202.17
Gainesvile	**0.19**	64.94	34.94	2.34	362.30	84.57	34.26	52.69	203.50
Honghu	**0.31**	112.51	57.30	4.45	585.11	132.43	47.22	151.33	211.48
Average	**0.23**	76.63	39.67	2.82	409.04	95.88	36.83	73.47	207.11

Besides, to evaluate the computational burden, as illustrated in Table V, we further tabulate the actual runtime of different methods. RX, CRD, LRASR, MoG, PAB_dc, PTA, and RGAE are performed using MATLAB R2020a on an Intel[1] Xeon[1] CPU E5-2620 v4 at 2.10 GHz and 64 GB of RAM. AE and the proposed MDMN are implemented by PyTorch on an NVIDIA 2080 Ti GPU with 11-GB memory. During the optimization, since forward propagation of multiple parallel AEs in **MixNet** is performed on a single device iteratively, the proposed method has no advantage in the computational cost. Future work will focus on developing a model-based parallelism learning scheme that partitions the proposed MDMN and deploys it on heterogeneous and distributed devices to

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deal with massive volumes of HSI and improve computational efficiency.

V. Conclusion

This article proposes a novel robust HAD framework using a union of AEs and a soft-membership assignment network, called the MDMN, from an interpretable perspective that incorporates the advantages of model-driven and data-driven approaches. It is based on the assumption that various background classes in HSIs with spectral variability and diverse structures are generated from a mixture of low-dimensional manifolds. To provide a comprehensive characterization of the background representation, we establish a novel deep mixture module. This module associates a single low-dimensional manifold learned by an AE with a certain background class. By leveraging the spectrum-specific sparsity prior of anomalies, we apply a sparse constraint to the anomaly component. This constraint serves to eliminate anomaly corruption and prevent the MDMN from overfitting to specific anomalies. While achieving excellent background representation, the MDMN enhances robustness and exhibits significant modeling deviations in the case of anomalies. Extensive experiments on various real datasets validate the effectiveness of our proposed algorithm through visual comparison and quantitative evaluation.

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